Lesson 57: Actor-Critic and Generalized Advantage Estimation

AI/ML Mathematics

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1 Motivation and Overview

REINFORCE (Lesson 55) uses the complete return $G_t = \sum_{k=0}^{T-t-1} \gamma^k r(S_{t+k}, A_{t+k})$ to estimate the advantage, yielding an unbiased but high-variance gradient. At the other extreme, a one-step temporal difference (TD) error $\delta_t = r(S_t, A_t) + \gamma \hat{V}(S_{t+1}) - \hat{V}(S_t)$ provides a low-variance but biased estimate of the advantage. Both are limiting cases of a family of estimators parameterized by a tradeoff parameter $\lambda \in [0, 1]$. Can we find the optimal point in this family?

This is exactly the question addressed by *Generalized Advantage Estimation* (GAE) of Schulman et al. [2016]. GAE defines a λ-weighted exponential average of multi-step advantage estimates, analogous to TD(λ) in value estimation (Lesson 24), but applied to the advantage function. The resulting estimator $\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)}$ provides a smooth knob between bias and variance: $\lambda = 0$ gives the one-step TD advantage (low variance, high bias), $\lambda = 1$ gives the Monte Carlo advantage (high variance, zero bias). The tools developed here form the backbone of modern policy gradient implementations including PPO [Schulman et al., 2017] and A3C [Mnih et al., 2016].

RL/ML Connection

- **A2C/A3C** [Mnih et al., 2016]: Uses actor-critic with n-step returns; GAE generalizes this to λ-weighted multi-step returns.
- **PPO** [Schulman et al., 2017]: Uses GAE to estimate advantages within the clipped surrogate objective, with $\lambda = 0.95$ as a common default.
- **Natural actor-critic** [Peters and Schaal, 2008]: The critic estimates the advantage using compatible features; GAE provides an alternative estimation strategy.
- **TRPO** [Schulman et al., 2015]: The original GAE paper combines GAE with trust-region optimization for policy updates.

Prerequisites. The reader must know the policy gradient theorem (Lesson 54, Theorem 5.1), the advantage function $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ and its role in policy gradients (Lesson 55, Definition 4.3 and Proposition 4.5), REINFORCE with baseline (Lesson 55, Section 4), and the compatible function approximation theorem (Lesson 54, Theorem 8.2). From the temporal difference methods lesson (Lesson 24), familiarity with TD(λ), eligibility traces, and the bias-variance tradeoff in bootstrapping is assumed. The natural policy gradient (Lesson 56) provides useful context but is not strictly required.

Learning objectives.

- **Define** the actor-critic architecture and explain the roles of the actor (policy) and critic (value function).
- **Prove** that the one-step TD error δ_t is an unbiased estimate of the advantage $A^\pi(S_t, A_t)$ when $\hat{V} = V^\pi$.
- **Derive** the n-step advantage estimator and analyze its bias and variance properties.
- **Derive** Generalized Advantage Estimation as an exponentially weighted average of n-step estimators.
- **Analyze** the bias-variance tradeoff of GAE as a function of λ.
- **Compute** GAE for concrete value function estimates on simple MDPs.

2 References

- Schulman et al. [2016] – The GAE paper; derives the λ-weighted advantage estimator with bias-variance analysis.
- Sutton and Barto [2018, Chapter 13] – Actor-critic methods, including one-step actor-critic and eligibility traces.
- Konda and Tsitsiklis [2003] – Rigorous two-timescale convergence analysis of actor-critic algorithms.
- Mnih et al. [2016] – A3C: asynchronous actor-critic with n-step returns in deep RL.
- Schulman et al. [2017] – PPO: practical use of GAE in large-scale policy optimization.
- Bhatnagar et al. [2009] – Convergence of natural actor-critic with TD-based critics.

3 The Actor-Critic Architecture

Actor-critic methods maintain two components: a parameterized policy (the actor) and a learned value function (the critic). The critic provides low-variance estimates of the advantage, which the actor uses to compute policy gradient updates.

Definition 3.1 (Actor-Critic Components)**.** An *actor-critic* system consists of:

(i) **Actor:** a parameterized policy $\pi_\theta(a \mid s)$ with parameters $\theta \in \mathbb{R}^d$.

(ii) **Critic:** a parameterized value function $\hat{V}_w(s)$ with parameters $w \in \mathbb{R}^m$, serving as an approximation to $V^\pi(s)$.

The actor is updated via policy gradient ascent using advantage estimates provided by the critic. The critic is updated to minimize the value prediction error, typically via temporal difference learning.

Definition 3.2 (One-Step Actor-Critic Update)**.** At each time step t, given the transition $(S_t, A_t, R_{t+1}, S_{t+1})$:

1. Compute the TD error: $\delta_t = R_{t+1} + \gamma\, \hat{V}_w(S_{t+1}) - \hat{V}_w(S_t)$.
2. Update the critic: $w \leftarrow w + \beta\, \delta_t\, \nabla_w \hat{V}_w(S_t)$.
3. Update the actor: $\theta \leftarrow \theta + \alpha\, \gamma^t\, \delta_t\, \nabla_\theta \log \pi_\theta(A_t \mid S_t)$.

RL/ML Connection

The one-step actor-critic is the simplest actor-critic algorithm. It replaces the full return G_t in REINFORCE with the TD error δ_t, which uses bootstrapping through $\hat{V}_w(S_{t+1})$. This introduces bias (since $\hat{V}_w \neq V^\pi$ in general) but dramatically reduces variance, since δ_t depends only on a single transition rather than the entire trajectory. The two-timescale design (with step sizes $\beta \gg \alpha$) allows the critic to track the value function of the slowly changing actor.

4 The TD Error as Advantage Estimate

The temporal difference error provides a natural connection between value function learning and advantage estimation.

Definition 4.1 (Temporal Difference Error)**.** Given a value function estimate $\hat{V}$, the *TD error* at time t is

$$\delta_t^{\hat{V}} = r(S_t, A_t) + \gamma\, \hat{V}(S_{t+1}) - \hat{V}(S_t).$$

When $\hat{V} = V^\pi$, we write δ_t^π and call it the *true TD error*.

Theorem 4.2 (TD Error as Unbiased Advantage Estimate)**.** *When $\hat{V} = V^\pi$, the TD error is an unbiased estimate of the advantage:*

$$\mathbb{E}[\delta_t^\pi \mid S_t = s, A_t = a] = A^\pi(s, a).$$

Proof. By definition and the Bellman equation:

$$\begin{aligned}
\mathbb{E}[\delta_t^\pi \mid S_t = s, A_t = a] &= \mathbb{E}[r(s,a) + \gamma\, V^\pi(S_{t+1}) - V^\pi(s) \mid S_t = s, A_t = a] \\
&= r(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a)\, V^\pi(s') - V^\pi(s) \\
&= Q^\pi(s,a) - V^\pi(s) = A^\pi(s,a),
\end{aligned}$$

where we used $Q^\pi(s,a) = r(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a)\, V^\pi(s')$. □

TD Error and the Advantage Function

The one-step TD error $\delta_t^\pi = r(S_t, A_t) + \gamma\, V^\pi(S_{t+1}) - V^\pi(S_t)$ provides an unbiased, single-sample estimate of $A^\pi(S_t, A_t)$ when the true value function is used. When an approximate value function $\hat{V}_w$ is used instead, the estimate becomes biased, with bias proportional to the value function approximation error.

Proposition 4.3 (Bias from Value Function Approximation)**.** *When $\hat{V}_w \neq V^\pi$, the TD error satisfies*

$$\mathbb{E}[\delta_t^{\hat{V}_w} \mid S_t = s, A_t = a] = A^\pi(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) \left(\hat{V}_w(s') - V^\pi(s')\right) - \left(\hat{V}_w(s) - V^\pi(s)\right).$$

The bias depends on the value function errors at s and at the successor states.

Proof. Adding and subtracting V^π: $\mathbb{E}[\delta_t^{\hat{V}_w} \mid s, a] = \underbrace{r(s,a) + \gamma \mathbb{E}[V^\pi(S_{t+1})] - V^\pi(s)}_{A^\pi(s,a)} + \gamma \mathbb{E}[\hat{V}_w(S_{t+1}) - V^\pi(S_{t+1})] - (\hat{V}_w(s) - V^\pi(s))$. □

Bias Cannot Be Eliminated by Baseline Invariance

Baseline invariance (Lesson 54, Proposition 7.1) guarantees that subtracting $V^\pi(s)$ from $Q^\pi(s,a)$ does not bias the gradient. However, when the TD target $r + \gamma \hat{V}_w(S_{t+1})$ replaces Q^π, the bootstrapping introduces bias that *cannot* be removed by baseline subtraction. This is fundamentally different from the baseline effect: the bias comes from using $\hat{V}_w(S_{t+1})$ to estimate future rewards, not from subtracting $\hat{V}_w(S_t)$.

5 Multi-Step Advantage Estimators

We now generalize the one-step TD advantage to n-step estimators, forming a family that interpolates between the TD(0) and Monte Carlo extremes.

Definition 5.1 (n-Step Advantage Estimator)**.** The *n-step advantage estimator* at time t is

$$\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k\, r(S_{t+k}, A_{t+k}) + \gamma^n\, \hat{V}(S_{t+n}) - \hat{V}(S_t) = -\hat{V}(S_t) + \sum_{k=0}^{n-1} \gamma^k\, r(S_{t+k}, A_{t+k}) + \gamma^n\, \hat{V}(S_{t+n}).$$

For $n = 1$, this recovers the TD error: $\hat{A}_t^{(1)} = \delta_t^{\hat{V}}$. As $n \to \infty$ (for episodic tasks, $n = T - t$), this becomes the Monte Carlo advantage: $\hat{A}_t^{(\infty)} = G_t - \hat{V}(S_t)$.

Proposition 5.2 (n-Step Estimator via TD Errors)**.** *The n-step advantage estimator can be written as a sum of discounted TD errors:*

$$\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k\, \delta_{t+k}^{\hat{V}}.$$

Proof. Expanding the TD errors:

$$\begin{aligned}\sum_{k=0}^{n-1} \gamma^k\, \delta_{t+k} &= \sum_{k=0}^{n-1} \gamma^k \left[r(S_{t+k}, A_{t+k}) + \gamma\, \hat{V}(S_{t+k+1}) - \hat{V}(S_{t+k})\right] \\ &= \sum_{k=0}^{n-1} \gamma^k\, r(S_{t+k}, A_{t+k}) + \sum_{k=0}^{n-1} \gamma^{k+1}\, \hat{V}(S_{t+k+1}) - \sum_{k=0}^{n-1} \gamma^k\, \hat{V}(S_{t+k}).\end{aligned}$$

The last two sums telescope: writing them out,

$$\begin{aligned}&\left[\gamma\, \hat{V}(S_{t+1}) + \gamma^2\, \hat{V}(S_{t+2}) + \cdots + \gamma^n\, \hat{V}(S_{t+n})\right] \\ &\quad - \left[\hat{V}(S_t) + \gamma\, \hat{V}(S_{t+1}) + \cdots + \gamma^{n-1}\, \hat{V}(S_{t+n-1})\right] = \gamma^n\, \hat{V}(S_{t+n}) - \hat{V}(S_t).\end{aligned}$$

Therefore $\sum_{k=0}^{n-1} \gamma^k \delta_{t+k} = \sum_{k=0}^{n-1} \gamma^k r(S_{t+k}, A_{t+k}) + \gamma^n \hat{V}(S_{t+n}) - \hat{V}(S_t) = \hat{A}_t^{(n)}$. □

Theorem 5.3 (Bias-Variance of n-Step Estimators)**.** *Assume $\hat{V} = V^\pi + \epsilon$ where $\epsilon(s)$ is the approximation error. Conditioning on $S_t = s$, $A_t = a$:*

(i) ***Bias:*** $\mathbb{E}[\hat{A}_t^{(n)} \mid s, a] = A^\pi(s, a) + O(\gamma^n\, \|\epsilon\|_\infty)$.

(ii) ***Variance:*** *The variance of $\hat{A}_t^{(n)}$ increases with n because more random transitions are included.*

Larger n reduces bias (the bootstrapped value $\hat{V}(S_{t+n})$ contributes less due to the γ^n discount) but increases variance (more stochastic transitions are summed).

Proof. (i) Writing $\hat{V} = V^\pi + \epsilon$, the expectation of the n-step estimator decomposes into $A^\pi(s, a)$ (from the true values) plus error terms $\gamma^n \mathbb{E}[\epsilon(S_{t+n})] - \epsilon(s) = O(\|\epsilon\|_\infty)$, with the γ^n factor shrinking as n grows.

(ii) Each additional term in $\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$ adds variance from the randomness of S_{t+k+1}, so total variance is non-decreasing in n. □

Example 5.4 (Comparing 1-Step and ∞-Step)**.** Consider a deterministic chain $S_0 \to S_1 \to \cdots \to S_T$ with $r_t = 0$ for $t < T - 1$, $r_{T-1} = 1$, $\gamma = 1$, $V^\pi(S_t) = 1$, and $\hat{V}(S_t) = 0.9$. The 1-step estimate $\hat{A}_0^{(1)} = 0 + 0.9 - 0.9 = 0$ misses the advantage (local errors cancel). The MC estimate $\hat{A}_0^{(\infty)} = 1 - 0.9 = 0.1$ correctly identifies positive advantage but carries full trajectory variance.

6 Generalized Advantage Estimation

GAE provides a principled way to interpolate between the n-step estimators by taking an exponentially weighted average.

Definition 6.1 (Generalized Advantage Estimation)**.** The *Generalized Advantage Estimator* GAE(γ, λ) at time t is the exponentially weighted average of n-step advantage estimators:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \, \hat{A}_t^{(n)},$$

where $\lambda \in [0, 1]$ controls the bias-variance tradeoff.

Theorem 6.2 (GAE as Discounted Sum of TD Errors)**.** *The GAE estimator admits the simple closed-form expression*

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \, \delta_{t+k}^{\hat{V}}.$$

For episodic tasks terminating at time T:

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{k=0}^{T-t-1} (\gamma\lambda)^k \, \delta_{t+k}^{\hat{V}}.$$

Proof. Substituting $\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k \, \delta_{t+k}$ (Proposition 5.2) into the GAE definition:

$$\begin{aligned}
\hat{A}_t^{\mathrm{GAE}} &= (1-\lambda) \sum_{n=1}^{\infty} \lambda^{n-1} \sum_{k=0}^{n-1} \gamma^k \, \delta_{t+k} \\
&= (1-\lambda) \sum_{k=0}^{\infty} \gamma^k \, \delta_{t+k} \sum_{n=k+1}^{\infty} \lambda^{n-1},
\end{aligned}$$

where we swapped the order of summation (justified by absolute convergence, since $|\delta_{t+k}| \leq R_{\max}/(1-\gamma)$ and $\gamma\lambda < 1$). The inner sum is:

$$\sum_{n=k+1}^{\infty} \lambda^{n-1} = \lambda^k \sum_{j=0}^{\infty} \lambda^j = \frac{\lambda^k}{1-\lambda}.$$

Therefore:

$$\hat{A}_t^{\mathrm{GAE}} = (1-\lambda) \sum_{k=0}^{\infty} \gamma^k \, \delta_{t+k} \, \frac{\lambda^k}{1-\lambda} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \, \delta_{t+k}.$$

□

Generalized Advantage Estimation

$$\hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \, \delta_{t+k}^{\hat{V}},$$

where $\delta_{t+k}^{\hat{V}} = r(S_{t+k}, A_{t+k}) + \gamma \, \hat{V}(S_{t+k+1}) - \hat{V}(S_{t+k})$. The parameter λ interpolates between:

- $\lambda = 0$: $\hat{A}_t^{\mathrm{GAE}} = \delta_t^{\hat{V}}$ (one-step TD advantage; low variance, high bias).
- $\lambda = 1$: $\hat{A}_t^{\mathrm{GAE}} = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} = G_t - \hat{V}(S_t)$ (Monte Carlo advantage; zero bias when $\hat{V} = V^\pi$, high variance).

Remark 6.3 (Recursive Computation of GAE). In practice, GAE is computed recursively backward through the trajectory:

$$\hat{A}_{T-1}^{\text{GAE}} = \delta_{T-1}, \qquad \hat{A}_t^{\text{GAE}} = \delta_t + \gamma\lambda\, \hat{A}_{t+1}^{\text{GAE}} \quad \text{for } t = T-2, \ldots, 0.$$

This follows directly from the geometric series structure and costs $O(T)$ time for an episode of length T.

7 Bias-Variance Analysis of GAE

We now formally analyze the bias and variance of GAE as a function of λ.

Theorem 7.1 (Bias of GAE). *Let $\hat{V} = V^\pi + \epsilon$ where $\epsilon : \mathcal{S} \to \mathbb{R}$ is the value function approximation error. Then:*

$$\mathbb{E}\Big[\hat{A}_t^{\text{GAE}(\gamma,\lambda)} \mid S_t = s, A_t = a\Big] = A^\pi(s,a) + \sum_{k=0}^{\infty} (\gamma\lambda)^k\, b_{t+k}(s,a),$$

where b_{t+k} represents the bias contribution from the k-th TD error, satisfying $|b_{t+k}(s,a)| \leq (1+\gamma)\|\epsilon\|_\infty$.

Proof. By Proposition 4.3, each TD error has bias $\mathbb{E}[\delta_{t+k}^{\hat{V}} \mid S_{t+k}, A_{t+k}] = A^\pi + \gamma\,\mathbb{E}[\epsilon(S_{t+k+1})] - \epsilon(S_{t+k})$. Summing with weights $(\gamma\lambda)^k$ and taking iterated expectations gives $\mathbb{E}[\hat{A}_t^{\text{GAE}} \mid s,a] = A^\pi(s,a) + \text{bias terms}$. For $\lambda < 1$, the total bias is bounded by $(1+\gamma)\|\epsilon\|_\infty/(1-\gamma\lambda)$. □

Corollary 7.2 (Unbiasedness at $\lambda = 1$). *When $\hat{V} = V^\pi$ (i.e., $\epsilon = 0$), $\hat{A}_t^{\text{GAE}(\gamma,\lambda)}$ is an unbiased estimate of $A^\pi(S_t, A_t)$ for* all $\lambda \in [0,1]$. *When $\hat{V} \neq V^\pi$, the estimator is unbiased only at $\lambda = 1$ (the Monte Carlo case).*

Proof. When $\epsilon = 0$, each TD error δ_{t+k}^π satisfies $\mathbb{E}[\delta_{t+k}^\pi \mid S_{t+k}, A_{t+k}] = A^\pi(S_{t+k}, A_{t+k})$ by Theorem 4.2. For $\lambda = 1$, $\hat{A}_t^{\text{GAE}(\gamma,1)} = \sum_{k=0}^{\infty} \gamma^k \delta_{t+k} = G_t - \hat{V}(S_t) = G_t - V^\pi(S_t)$, which is an unbiased estimate of $Q^\pi(S_t, A_t) - V^\pi(S_t) = A^\pi(S_t, A_t)$ by Lemma 3.2 of Lesson 55.

For $\lambda < 1$ and $\epsilon \neq 0$, the bias is $O(\|\epsilon\|_\infty/(1-\gamma\lambda)) \neq 0$ in general. □

Theorem 7.3 (Variance of GAE). *Consider the simplified setting where the TD errors $\delta_0, \delta_1, \ldots$ are uncorrelated (which holds when $\hat{V} = V^\pi$ and the MDP is such that each δ_t depends only on the local transition). Then the variance of GAE is:*

$$\text{Var}\Big(\hat{A}_t^{\text{GAE}(\gamma,\lambda)}\Big) = \sum_{k=0}^{\infty} (\gamma\lambda)^{2k}\, \text{Var}(\delta_{t+k}^\pi).$$

If additionally $\text{Var}(\delta_{t+k}^\pi) \leq \sigma_\delta^2$ for all k, then

$$\text{Var}\Big(\hat{A}_t^{\text{GAE}}\Big) \leq \frac{\sigma_\delta^2}{1-\gamma^2\lambda^2}.$$

Proof. Under the uncorrelatedness assumption:

$$\text{Var}\left(\sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}\right) = \sum_{k=0}^{\infty} (\gamma\lambda)^{2k}\, \text{Var}(\delta_{t+k}).$$

Bounding each $\text{Var}(\delta_{t+k}) \leq \sigma_\delta^2$ and summing the geometric series gives the result. □

RL/ML Connection

The bias-variance tradeoff is the central design choice in GAE:

- **Small** λ (e.g., $\lambda = 0$): Variance $\approx \sigma_\delta^2$, bias $\approx (1+\gamma)\|\epsilon\|_\infty/(1-\gamma\lambda)$. Good when the critic is accurate ($\|\epsilon\|$ small).
- **Large** λ (e.g., $\lambda = 0.95$): Variance $\approx \sigma_\delta^2/(1-\gamma^2\lambda^2)$, bias is small even with inaccurate critic. This is the typical choice in practice (PPO uses $\lambda = 0.95$ by default).
- $\lambda = 1$: Zero bias, maximum variance. Equivalent to REINFORCE with baseline.

8 The GAE Policy Gradient

We now state the policy gradient estimator that uses GAE advantages and analyze its properties.

Definition 8.1 (GAE Policy Gradient Estimator)**.** Given a trajectory $(S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T)$ collected under π_θ, the *GAE policy gradient estimator* is

$$\hat{g}^{\text{GAE}} = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, \hat{A}_t^{\text{GAE}(\gamma,\lambda)}.$$

Proposition 8.2 (Bias of the GAE Policy Gradient)**.** *When $\hat{V} = V^\pi$, $\hat{g}^{\text{GAE}}$ is unbiased for all λ (by Corollary 7.2). When $\hat{V} \neq V^\pi$ and $\lambda < 1$, the estimator is biased (Theorem 7.1). When $\lambda = 1$, $\hat{A}_t^{\text{GAE}} = G_t - \hat{V}(S_t)$ and $\hat{V}(S_t)$ acts as a state-dependent baseline, so the estimator is unbiased by baseline invariance (Lesson 54, Proposition 7.1) regardless of $\hat{V}$.*

Example 8.3 (GAE Computation on a Simple Trajectory)**.** Consider a trajectory of length $T = 4$ with $\gamma = 0.99$, $\lambda = 0.95$, and the following data:

t	r_t	$\hat{V}(S_t)$	$\hat{V}(S_{t+1})$	δ_t
0	1.0	10.0	8.0	$1.0 + 0.99 \times 8.0 - 10.0 = -1.08$
1	2.0	8.0	6.0	$2.0 + 0.99 \times 6.0 - 8.0 = -0.06$
2	3.0	6.0	5.0	$3.0 + 0.99 \times 5.0 - 6.0 = 1.95$
3	4.0	5.0	0.0	$4.0 + 0.99 \times 0.0 - 5.0 = -1.00$

Computing GAE recursively backward (Remark 6.3) with $\gamma\lambda = 0.99 \times 0.95 = 0.9405$:

$$\begin{aligned}
\hat{A}_3^{\text{GAE}} &= \delta_3 = -1.00, \\
\hat{A}_2^{\text{GAE}} &= \delta_2 + 0.9405 \times (-1.00) = 1.95 - 0.94 = 1.01, \\
\hat{A}_1^{\text{GAE}} &= \delta_1 + 0.9405 \times 1.01 = -0.06 + 0.95 = 0.89, \\
\hat{A}_0^{\text{GAE}} &= \delta_0 + 0.9405 \times 0.89 = -1.08 + 0.84 = -0.24.
\end{aligned}$$

For comparison, with $\lambda = 0$ (TD-only): $\hat{A}_t = \delta_t$, so $\hat{A}_0 = -1.08$. With $\lambda = 1$ (Monte Carlo): $\hat{A}_0 = (1 + 0.99 \times 2 + 0.99^2 \times 3 + 0.99^3 \times 4) - 10.0 = 9.80 - 10.0 = -0.20$. The GAE estimate ($-0.24$) lies between the TD ($-1.08$) and MC ($-0.20$) values, as expected.

9 Two-Timescale Convergence of Actor-Critic

Assumption 9.1 (Two-Timescale Step Sizes)**.** The actor step sizes $\{\alpha_k\}$ and critic step sizes $\{\beta_k\}$ satisfy: (i) $\sum_k \alpha_k = \sum_k \beta_k = \infty$ and $\sum_k \alpha_k^2, \sum_k \beta_k^2 < \infty$; (ii) $\alpha_k/\beta_k \to 0$ (the actor updates on a slower timescale).

Theorem 9.2 (Two-Timescale Convergence)**.** *Under Assumption 9.1 and standard regularity conditions (Lipschitz gradients, bounded iterates, ergodicity), the critic w_k converges to $w^*(\theta)$ (the TD fixed point) and the actor θ_k converges to a stationary point of $J(\theta)$.*

Proof sketch. Since $\alpha_k/\beta_k \to 0$, the critic sees the actor as approximately constant and converges by standard TD theory (Lesson 24). The actor then uses approximately correct gradient estimates. The formal proof [Konda and Tsitsiklis, 2003] uses the ODE method: coupled stochastic iterations track a pair of ODEs, with the fast ODE (critic) converging on the slow timescale of the actor. □

RL/ML Connection

In practice, the strict two-timescale requirement is often relaxed. PPO and A3C use shared neural network layers for actor and critic, with the critic loss coefficient (e.g., $c_v = 0.5$) implicitly controlling the timescale ratio.

10 GAE with Policy Gradient Methods

GAE can be combined with any policy gradient method by replacing the advantage estimate. The vanilla policy gradient with GAE uses:

$$\theta_{k+1} = \theta_k + \alpha \sum_{t=0}^{T-1} \nabla_\theta \log \pi_{\theta_k}(A_t \mid S_t)\, \hat{A}_t^{\mathrm{GAE}(\gamma,\lambda)}.$$

The natural policy gradient with GAE (Lesson 56) instead uses $\theta_{k+1} = \theta_k + \alpha\, F(\theta_k)^{-1}\hat{g}$ where $\hat{g}$ is the GAE gradient estimate above.

Example 10.1 (Effect of λ on Credit Assignment)**.** In a grid-world with sparse rewards (+1 at the goal, 0 elsewhere): $\lambda = 0$ provides signal only one step from the goal (slow learning); $\lambda = 1$ propagates reward to all visited states (fast but noisy); $\lambda = 0.95$ propagates ~ 20 effective steps, balancing speed and stability.

Two Discount Parameters in GAE

GAE has *two* discount-like parameters: γ (the MDP discount defining the objective) and λ (controlling the bias-variance tradeoff). Setting $\lambda = 0.95$ with $\gamma = 0.99$ gives an effective estimation discount of $\gamma\lambda = 0.9405$. A common mistake is to confuse the two or set both to 1.

11 Exercises

Exercise 11.1 (TD Error Properties)**.** (∗)
For an MDP with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a\}$ (single action), deterministic transitions $s_1 \to s_2 \to s_1$, rewards $r(s_1, a) = 2$, $r(s_2, a) = 0$, and $\gamma = 0.9$:

(a) Compute $V^\pi(s_1)$ and $V^\pi(s_2)$ exactly.

(b) Verify that $\mathbb{E}[\delta_t^\pi \mid S_t = s_i]$ equals zero for both states (since there is only one action, the advantage is zero).

(c) If $\hat{V}(s_1) = 9$ and $\hat{V}(s_2) = 8$, compute the TD errors and the bias at each state.

Exercise 11.2 (GAE Recursive Computation). $(*)$
Verify the recursive formula $\hat{A}_t^{\text{GAE}} = \delta_t + \gamma\lambda\,\hat{A}_{t+1}^{\text{GAE}}$ by expanding the closed-form expression from Theorem 6.2. Then compute $\hat{A}_t^{\text{GAE}}$ for $t = 0, 1, 2$ on the trajectory $(r_0, r_1, r_2) = (1, 0, 5)$ with $\hat{V}(S_0) = 3$, $\hat{V}(S_1) = 2$, $\hat{V}(S_2) = 6$, $\hat{V}(S_3) = 0$ (terminal), $\gamma = 0.99$, $\lambda = 0.9$.

Exercise 11.3 (Bias-Variance Comparison). $(**)$
Consider a two-state MDP where $S_{t+1} \sim \text{Bernoulli}(0.5)$ independently, with $r(s, a) = s$ (the reward equals the state, treating $s \in \{0, 1\}$) and $\gamma = 0.9$. Suppose $\hat{V}(0) = \hat{V}(1) = 0.5$ and $V^\pi(0) = V^\pi(1) = 5.0$.

(a) Compute $\mathbb{E}[\hat{A}_0^{(1)}]$ and $\text{Var}(\hat{A}_0^{(1)})$ exactly.

(b) Compute $\mathbb{E}[\hat{A}_0^{(2)}]$ and show that the bias decreases with n.

(c) For $\lambda = 0.5$, compute $\mathbb{E}[\hat{A}_0^{\text{GAE}}]$ and compare with parts (a) and (b).

Exercise 11.4 (λ-Return Equivalence). $(**)$
The λ-*return* from TD(λ) is defined as $G_t^\lambda = (1-\lambda)\sum_{n=1}^{\infty} \lambda^{n-1}\, G_t^{(n)}$ where $G_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k\, r_{t+k} + \gamma^n\, \hat{V}(S_{t+n})$. Show that

$$\hat{A}_t^{\text{GAE}(\gamma,\lambda)} = G_t^\lambda - \hat{V}(S_t),$$

establishing that GAE is equivalent to using the λ-return as the target for the advantage, with $\hat{V}(S_t)$ as the baseline.

Exercise 11.5 (Optimal λ Selection). $(***)$
Consider the idealized setting where $\hat{V}(s) = V^\pi(s) + \epsilon$ with ϵ a constant (uniform bias). Show that in this case, the bias of $\hat{A}_t^{\text{GAE}(\gamma,\lambda)}$ is $\epsilon(\gamma\lambda - 1)/(1 - \gamma\lambda)$ for $\lambda < 1$ and zero for $\lambda = 1$. Then derive the mean-squared error MSE $=$ bias^2 $+$ variance as a function of λ (using the variance bound from Theorem 7.3) and find the λ^* that minimizes it as a function of ϵ and σ_δ.

Hint: The MSE-optimal λ balances $\partial(\text{bias}^2)/\partial\lambda$ against $\partial(\text{variance})/\partial\lambda$.

12 Summary

- The **actor-critic architecture** separates policy optimization (actor) from value function estimation (critic), allowing the critic to provide low-variance advantage estimates through bootstrapping.
- The **TD error** $\delta_t = r_t + \gamma\hat{V}(S_{t+1}) - \hat{V}(S_t)$ is an unbiased estimate of the advantage $A^\pi(S_t, A_t)$ when $\hat{V} = V^\pi$, but introduces bias when $\hat{V} \neq V^\pi$.
- The **n-step advantage estimator** $\hat{A}_t^{(n)} = \sum_{k=0}^{n-1} \gamma^k \delta_{t+k}$ interpolates between the TD error ($n = 1$) and the Monte Carlo advantage ($n = \infty$), with increasing n reducing bias but increasing variance.
- **Generalized Advantage Estimation** defines $\hat{A}_t^{\text{GAE}(\gamma,\lambda)} = \sum_{k=0}^{\infty} (\gamma\lambda)^k \delta_{t+k}$, providing a smooth bias-variance tradeoff via $\lambda \in [0, 1]$.
- At $\lambda = 0$, GAE reduces to the one-step **TD advantage** (low variance, high bias). At $\lambda = 1$, it becomes the **Monte Carlo advantage** (zero bias, high variance). Typical implementations use $\lambda = 0.95$.
- GAE is equivalent to using the **λ-return** G_t^λ as the advantage target with the value function as baseline, connecting policy gradient methods to the TD(λ) framework.

- **Two-timescale convergence**: When the critic learns faster than the actor ($\alpha_k/\beta_k \to 0$), the coupled system converges to a stationary point of $J(\theta)$.
- GAE can be combined with **any policy gradient method** (vanilla, natural, trust-region) by replacing the advantage estimate in the gradient expression.

13 Further Reading

- **Lesson 58** synthesizes policy gradient theory from Lessons 54–57, connecting to trust-region methods of Phase 04.
- Schulman et al. [2016] provides experiments showing the effect of λ on learning speed across continuous control benchmarks.
- Konda and Tsitsiklis [2003] gives the definitive two-timescale convergence proof for actor-critic methods.
- Mnih et al. [2016] demonstrates n-step actor-critic in A3C with asynchronous parallel workers.
- Schulman et al. [2017] uses GAE within PPO, showing $\lambda = 0.95$–0.99 works well across continuous control tasks.
- Sutton and Barto [2018, Chapter 12] develops TD(λ), from which GAE draws inspiration.

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